

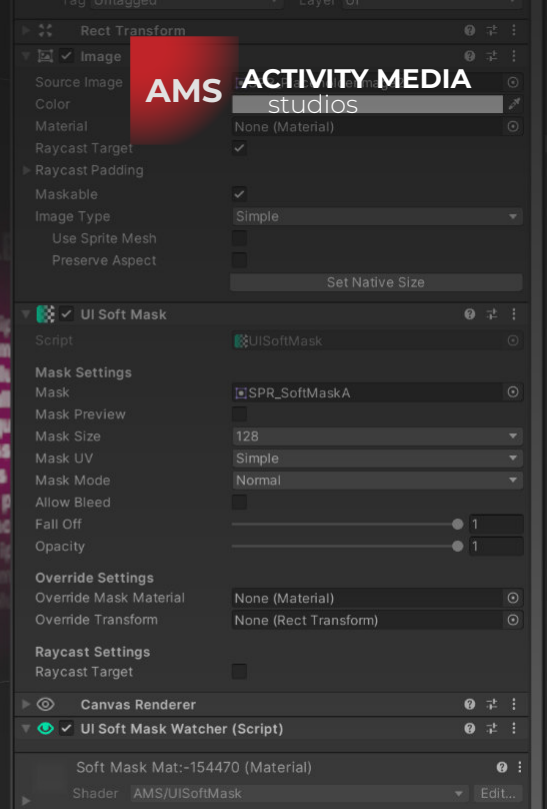


Unity Asset Package



UI Soft Mask

Documentation



1. Installation
2. Using AMS UI Soft Mask
 - 2.1 Properties
3. Custom Shader Support
 - 3.1 Shader Graph
 - 3.2 Amplify Shader
 - 3.3 Hand-coded shader

1. Installation

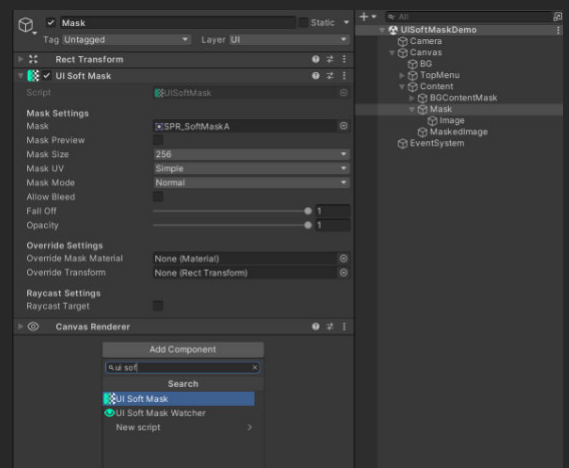
- Import 'AMS UI Soft Mask' through "Package Manager/My Assets"; or if you have a downloaded package import it as a custom package through "[Assets/Import Package/Custom Package](#)";
- It's required to have Unity's **TextMeshPro** package imported into your project;
- For **TextMeshPro shaders support** please import the package present in the plugin folder at path: "[Resources/Packages/TMP_SoftMaskSupport.unitypackage](#)".
Note: For different Unity versions consider adding TMP shader support manually. Please follow 3.3 Hand-coded Shader for that.

Important Note:

To ensure AMS UI Soft Mask shaders are included in builds make sure to add them to the 'Always Included Shaders' list at the menu "[Project Settings > Graphics > Always Included Shaders](#)". For your convenience, we've added a menu item at "Window/AMS/UISoftMask/Force Include Shaders (Project Settings)".

2. Using AMS UI Soft Mask

- Add a new "UI Soft Mask" component into your parent UI object;
- Hook your sprite mask into the "Mask" property;
- Control mask fall off by changing "Fall off" property;
- Control opacity by changing the "Opacity" property;



2.1 Properties

PROPERTY	DETAILS
Mask	A sprite asset that represents the mask for the hierarchy. Note: texture import settings require an alpha source (Input Texture Alpha or From Grayscale);
Mask Preview	Preview mask output. (Editor-only)
Mask Size	The texture size. Keep low to save memory allocation.
Mask UV	The mask UV's type. Choose between Simple and Sliced (9-slice masks).
Mask Mode	Choose between Normal (mask as drawn) or Inverted (flip the mask).
Allow Bleed	Allows mask bleeding outside the rect area. Note: It draws outside by stretching the rect pixel from each corresponding side. Use accordingly, as this behaviour may not be suitable for all mask setups.
Fall Off	Smooth step mask falloff control.
Opacity	Overall opacity control.
Override Mask Material	Use this to override the temporary UISoftMask material with a material asset from your project. Note: It requires a unique material per mask, and the shader must be compatible with this plugin.
Override Transform	Use this to override the mask transformation's size, position, and rotation to the specified tranform.
Raycast Target	When enabled, raycasts are filtered by the soft mask alpha, allowing to define a minimum threshold below which clicks will be ignored.

Important :

A watcher component ([UISoftMaskWatcher](#)) is required across the entire mask hierarchy to ensure proper material setup.

- It uses 'HideFlags.DontSaveInEditor' and 'HideFlags.DontSaveInBuild', meaning it won't be saved in prefabs.
- The component is automatically added or removed when a mask is used. If a maskable component is added to an existing object without hierarchy changes, a watcher must be added alongside it. Revalidation only occurs on start or hierarchy change.
- For implementation specifics, refer directly to the script source.

Note:

- If any maskable UI element present in the hierarchy uses a material with a custom shader, it is required to add 'UI Soft Mask' support to that shader.
- Please check section 3. Custom Shader Support for more details.

Tip:

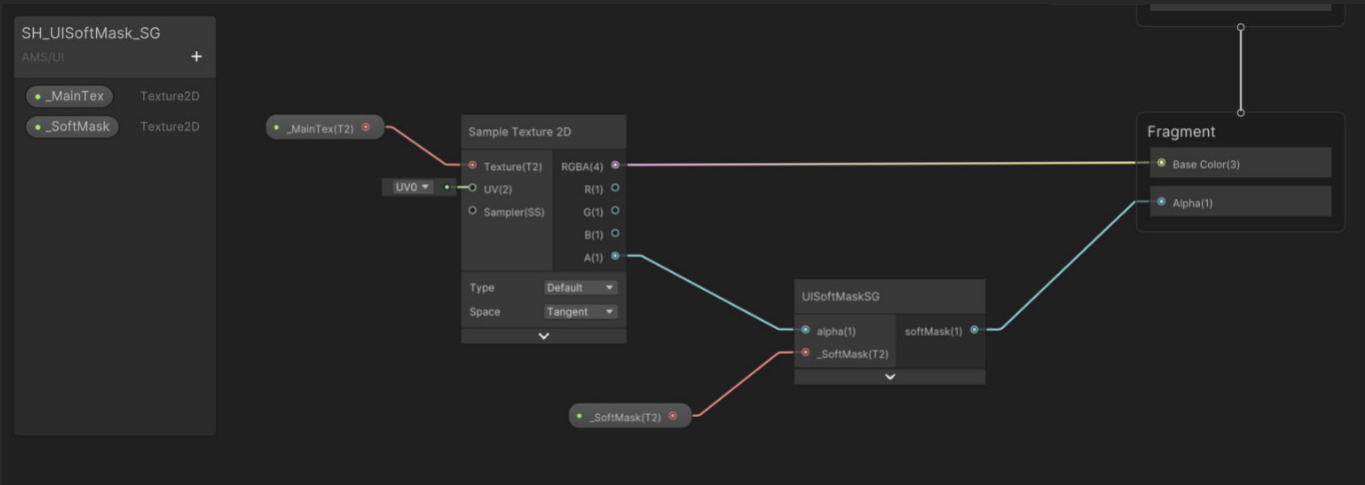
Use the "Override Mask Material" property to override the UISoftMask material in favor of using a target material. Note: It requires a unique material per mask and the shader must be compatible with AMS UI Soft Mask.

3. Custom Shader Support

AMS UI Soft Mask has support for custom shaders. In this section, you can find all the details about adding a soft mask to your UI custom shader.

3.1 Shader Graph

Into Shader Graph Editor search by "[UISoftMask](#)" to use the soft mask SubGraph like below:

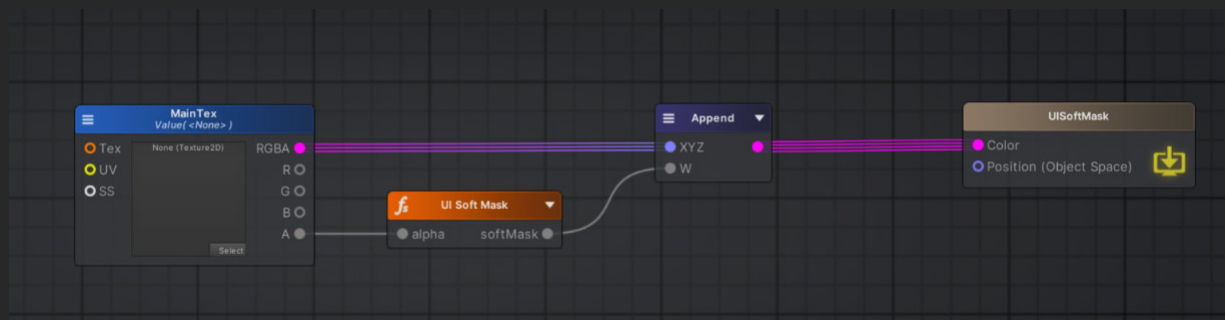


Notes:

1. If your Shader Graph version supports "Canvas Material Type", make sure to select it in "Graph Settings". Otherwise, you might experience unusual behaviour. If that's the case, we strongly recommend using a different shader editor or hand-coding shaders;
2. Please note that Shader Graph requires manually adding "_SoftMask (Texture2D)" property.

3.2 Amplify Shader Editor

Into Amplify Shader Editor search by "[UI Soft Mask](#)" to use the soft mask Amplify Shader Function like below:



3.3 Hand-coded Shader

If you are using hand-coded shaders for the UI, please add the following lines of code to your shader in their respective locations.

```
Shader "Example/UISoftMask 3.0"
{
    Properties
    {
        //...
        // 1. SHADER PROPERTIES
        // Add this line into ShaderLab Properties Block
        [PerRendererData]_SoftMask("_SoftMask", 2D) = "white" {}
        //...
    }
    SubShader
    {
        Pass
        {
            CGPROGRAM
            //...
            // 2. INCLUDE & PROPERTIES
            // Add these lines before your 'Vertex Output' struct.
            // Note: Make sure to change include path if you are using a different location for 'AMS UI Soft Mask' package
            #pragma multi_compile_local _ USING_SOFT_MASK
            #include_with_pragmas "Assets/AMS/UISoftMask/Shader/Utils/UISoftMask.hlsl"
            uniform sampler2D _SoftMask;
            //...

            struct Varyings
            {
                //...
                // 3. VERTEX OUTPUT STRUCT
                // Add these lines into 'Vertex Output' struct.
                #ifdef USING_SOFT_MASK
                float2 softMaskUV : TEXCOORDX; // 'X' being the TexCoord Index available on struct
                #endif
                //...
            }

            Varyings vert (Attributes IN)
            {
                //...
                // 4. VERTEX FUNCTION
                // Add these lines into Vertex function
                // Note: 'IN.vertex' being the corresponding 'Vertex Data : POSITION' and 'OUT' the 'Vertex Output' in vertex function
                #ifdef USING_SOFT_MASK
                float4 wPos = mul(unity_ObjectToWorld, IN.vertex);
                OUT.softMaskUV = MaskUV(wPos);
                #endif
                //...
            }

            half4 frag(Varyings IN) : SV_Target
            {
                //...
                // 5. FRAGMENT FUNCTION
                // Add these lines into Fragment function just before UNITY_UI_ALPHACLIP instruction
                // Note: 'IN' being the corresponding 'Vertex Output' and 'color' the candidate fragment output
                #ifdef USING_SOFT_MASK
                color.a = UISoftMask(IN.softMaskUV, _SoftMask, color.a);
                #endif
                //...
            }
            ENDCG
        }
    }
}
```